

CSE471 — DEEP LEARNING FOR COMPUTER VISION

Dataset Considerations in Computer Vision

Small Data • Class Imbalance • Dataset Splitting

Week 7 | Lecture 14

Unit III: Transfer Learning



Unit III Progression: Where We Are



Core Question: What if our model strategy is perfect, but our dataset is the real problem?

Learning Objectives

-  **Small Data Challenges:** Understand why dataset size matters, the risk of overfitting, and how data augmentation helps.
-  **Class Imbalance:** Explain why severe imbalance occurs, why standard accuracy metrics can be highly misleading, and how to detect it.
-  **Dataset Splitting:** Understand the critical difference between Training, Validation, and Test sets, and how to avoid data leakage.
-  **Practical Application:** Apply these concepts directly to the upcoming Cats vs. Dogs fine-tuning capstone project.

How Much Data Does a Deep Learning Model Need?

Can a deep neural network learn from only 100 images?



100 Images

Extremely challenging. High risk of the model simply memorizing the exact pixels rather than learning general concepts.



10,000 Images

Often sufficient for specific tasks, especially when combined with transfer learning from a pretrained base.



1,000,000+ Images

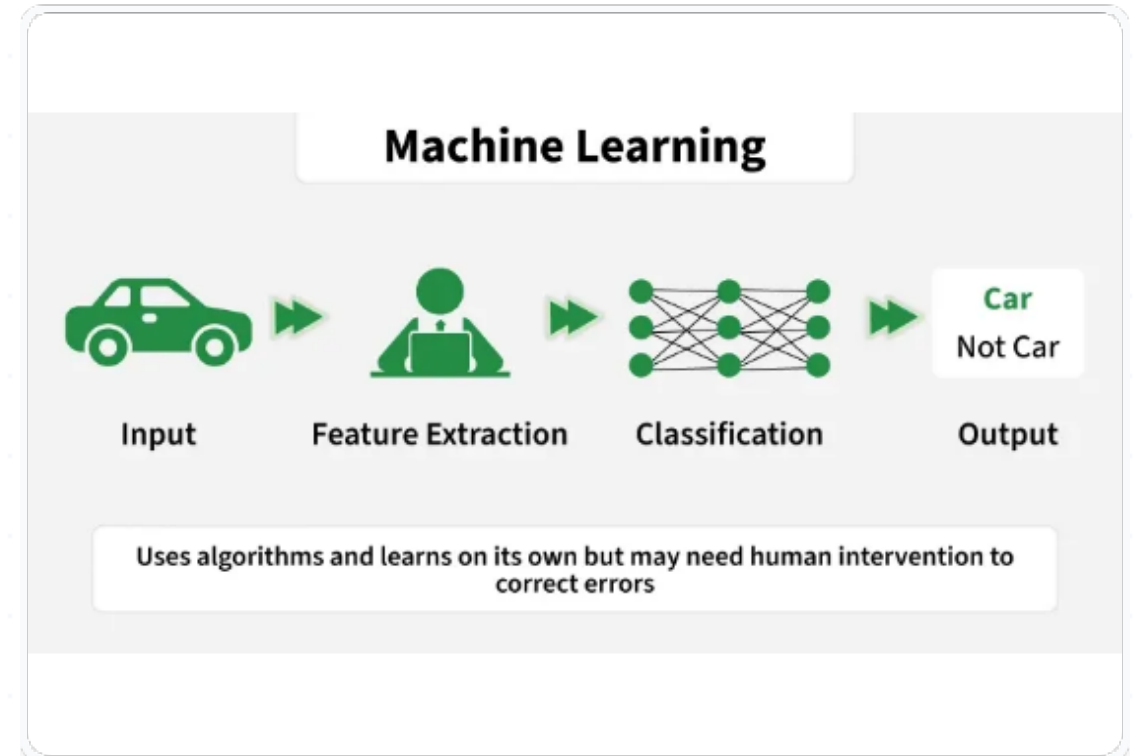
Standard for foundational models (e.g., ImageNet). Allows the network to learn rich, robust hierarchical features.

Note: There is no universal "magic number" that guarantees success.

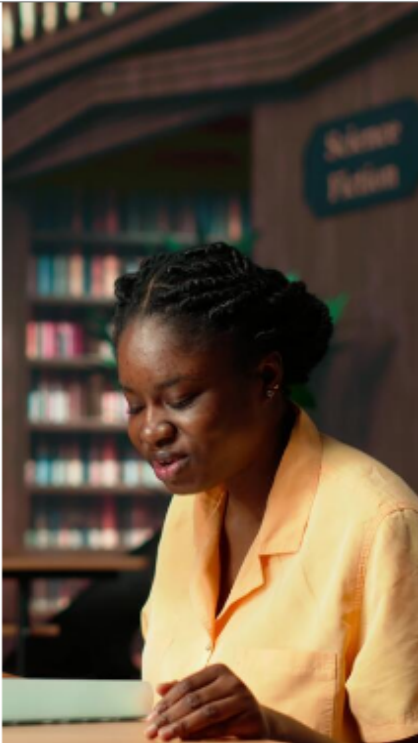
Why Is Small Data Difficult?

- **Limited Exposure:** The model sees fewer examples and may miss rare but important patterns.
- **Memorization Risk:** Deep neural networks have millions of parameters; they can easily memorize a tiny dataset.
- **Poor Generalization:** It may fail entirely on unseen images that look slightly different.
- **Unreliable Evaluation:** Testing on a small set of images doesn't prove the model works reliably in the real world.

→ This fundamental problem leads us to our first major concept: Overfitting.



Understanding Overfitting



Definition

Overfitting happens when a model learns the training data *too specifically* and consequently performs poorly on new, unseen data.

The Analogy

Imagine a student who memorizes the exact answers to a previous year's exam paper.

- **Training phase:** The student scores 100% on the practice test.
- **Testing phase:** The actual exam has slightly different questions. The student fails because they memorized, rather than learning the underlying concepts.

Training vs. Generalization Performance



Notice how the model continues to get better at the training data, while simultaneously getting *worse* at generalizing to new, unseen data.

Strategies for Small Datasets



Transfer Learning

Use pretrained weights. They already "know" basic visual features, reducing the need for massive data.



Data Augmentation

Artificially expand your dataset by modifying existing images.



Regularization

Techniques like Dropout that mathematically prevent the network from memorizing.



Simpler Models

A smaller model has fewer parameters, making it inherently harder for it to overfit.



Collect More Data

When possible, nothing beats collecting more high-quality real-world examples.

Data Augmentation

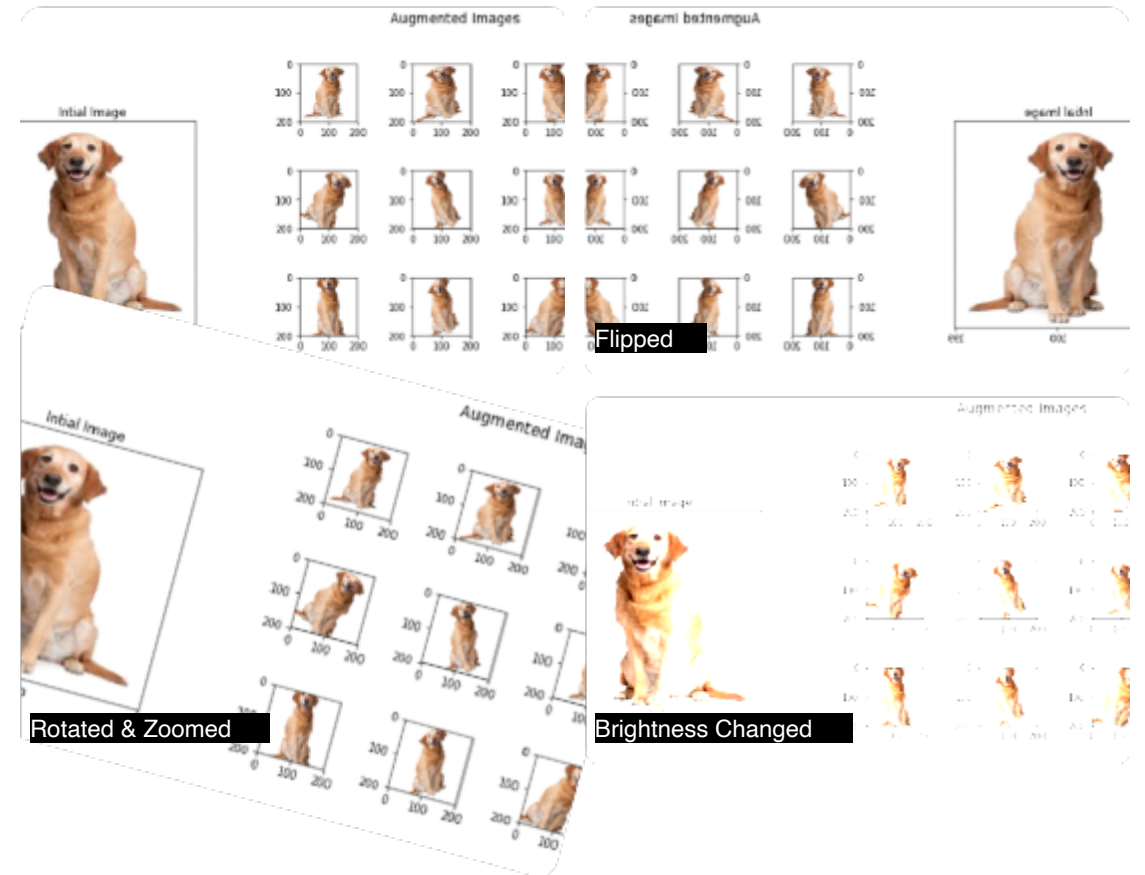
Definition

Data augmentation creates modified versions of existing training images to increase the variety of examples available to the model.

It acts like a multiplier for your dataset, forcing the model to learn robust features rather than specific pixel locations.

Common Transformations:

- Horizontal / Vertical Flip
- Slight Rotations
- Random Crops
- Brightness / Contrast Changes
- Slight Zoom



Good vs. Bad Augmentation

✓ Usually Reasonable

- Horizontal flip (for objects like animals/cars).
- Small rotations (± 15 degrees).
- Random cropping to focus on parts.
- Slight lighting and color changes.

Goal: Create realistic variations that occur in the real world.

✗ Potentially Problematic

- Extreme rotation (e.g., upside down) if context matters.
- Unrealistic distortion or warping.
- Color inversion for natural images.
- Changes that alter the object's meaning.



Flipping a '60' sign makes it meaningless.

Rule of Thumb: Augmentation should preserve the correct label.

Part 2: What Is Class Imbalance?

Class imbalance occurs when some categories in your dataset have significantly more examples than others.

Class A (Majority)



90% of Dataset

Class B (Minority)



10% of Dataset

Result: The model sees far more examples of Class A, biasing its learning.

Real-World Example: Defective Products

Suppose an automated factory assembly line dataset contains 10,000 images:

- 9,900 Normal products
- 100 Defective products

A model trains on this and learns to predict "**Normal**" for every single image it ever sees.

Accuracy: 99%



Why Accuracy Can Be Misleading

99%

Accuracy

The Illusion of Performance

Accuracy asks a simple question: *"How many predictions were correct overall?"*

With severe imbalance, a model can perform perfectly mathematically on the majority class, while completely failing on the critical minority class.

We need better evaluation metrics:

- **Precision**
- **Recall**
- **F1-Score**
- **Confusion Matrix**

Confusion Matrix Intuition

A simple 2x2 grid that reveals exactly *where* the model is making mistakes.

	Predicted Defect (Positive)	Predicted Normal (Negative)
Actual Defect (Positive)	True Positive (TP) Correctly caught defect	False Negative (FN) Missed the defect (Costly!)
Actual Normal (Negative)	False Positive (FP) False alarm	True Negative (TN) Correctly identified normal

Our 99% accuracy model had 0 True Positives and 100 False Negatives!

Handling Class Imbalance



Data-Level Approaches

Collect more examples of the minority class (if possible). Use *oversampling* or targeted data augmentation to artificially increase minority representation.



Model/Training-Level Approaches

Apply *Class Weights* during training. This forces the model's loss function to penalize errors on the minority class much more heavily than errors on the majority class.



Evaluation-Level Approaches

Never rely solely on Accuracy. Always inspect the confusion matrix to ensure both classes are being learned effectively.

Oversampling vs. Undersampling

Oversampling

Increase the number of examples in the minority class (e.g., by duplicating them or using augmentation).

Majority: ●●●●
Minority: ● → ●●●●

Note: Duplicating doesn't create new information, risk of overfitting the minority class.

Undersampling

Reduce the number of examples in the majority class by randomly removing them.

Majority: ●●●● → ●
Minority: ●

Note: You are throwing away perfectly good data, which networks usually need.

Part 3: Dataset Splitting

“ If we train our model and test its accuracy using exactly the same images, what does the test accuracy actually tell us? ”

Absolutely nothing about its real-world performance.

The Three Splits: Train / Validation / Test

Entire Dataset (100%)



1. Training Set

The data the model actually "sees" and learns from to update its internal weights/parameters.

2. Validation Set

Used to evaluate performance *during* development. Helps tune hyperparameters and detect overfitting.

3. Test Set

Locked away completely. Used exactly **once** at the very end to evaluate final, objective performance.

Typical Split Ratios



Standard / Small Datasets

Common when dealing with thousands of images. Ensures enough data for validation and testing.



Common Deep Learning

Typical starting point. Maximizes training data while keeping a reasonable evaluation buffer.

Note: These are guidelines, not universal rules. With massive datasets (millions of images), splits like 98% / 1% / 1% are common because 1% is still tens of thousands of images.

The Golden Rule: Avoid Data Leakage

Definition

Data Leakage occurs when information from outside the training process improperly influences model training or evaluation.

✗ **Test Data → Training**

The Correct Flow

- ✓ **Training** → Updates Model
- ✓ **Validation** → Model Selection
- ✓ **Test** → Final Evaluation Only

Applying Data Augmentation to your Validation or Test sets is a common form of leakage!

A Common Image-Splitting Mistake

Suppose you have a dataset of faces, including a "burst" of 10 photos of the exact same person taken 1 second apart.

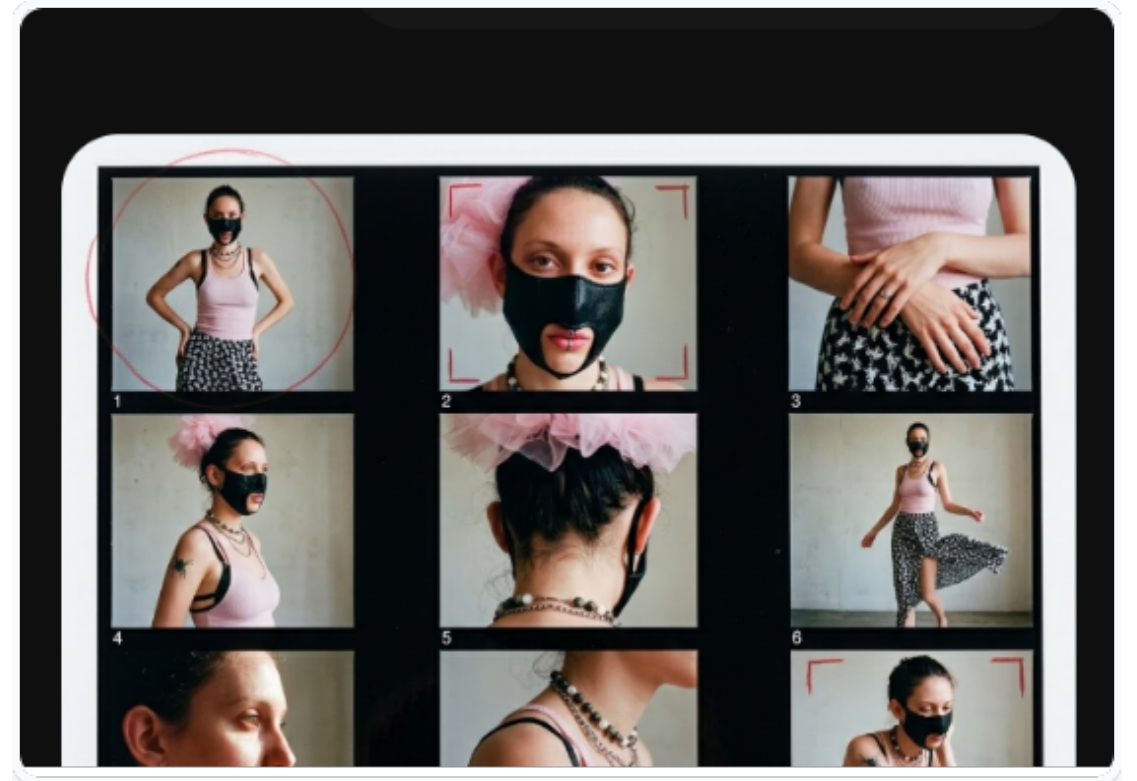
If you perform a completely **Random Split**:

- Photo 1, 2, 4, 7 → Training Set
- Photo 3, 9 → Test Set

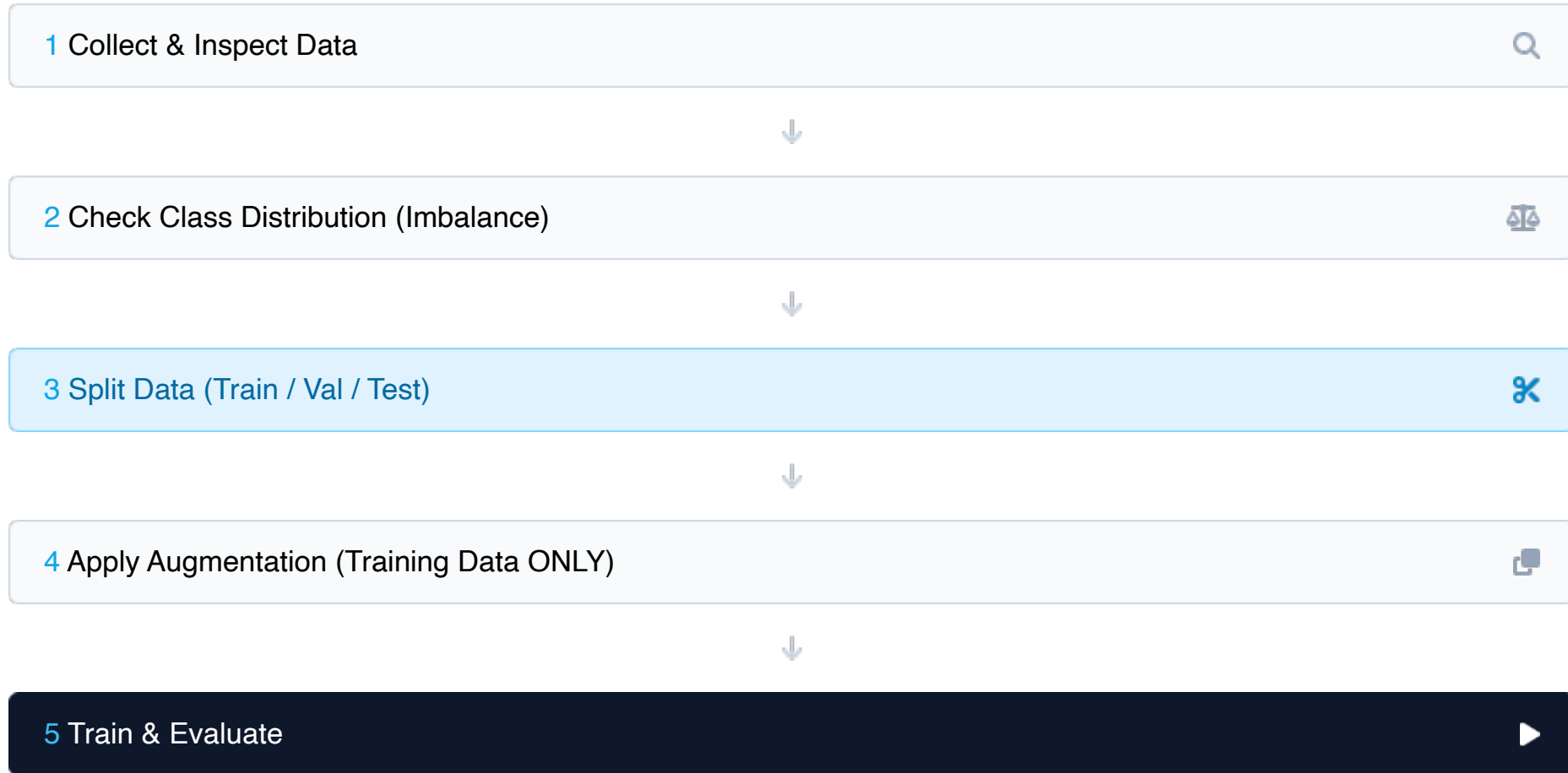
Result: Artificially High Test Performance

The model has practically already seen the test images during training.

Solution: Group images by subject/video clip before splitting.



Dataset Preparation Workflow



Practical 6 — Fine-Tuning MobileNet on Cats vs. Dogs

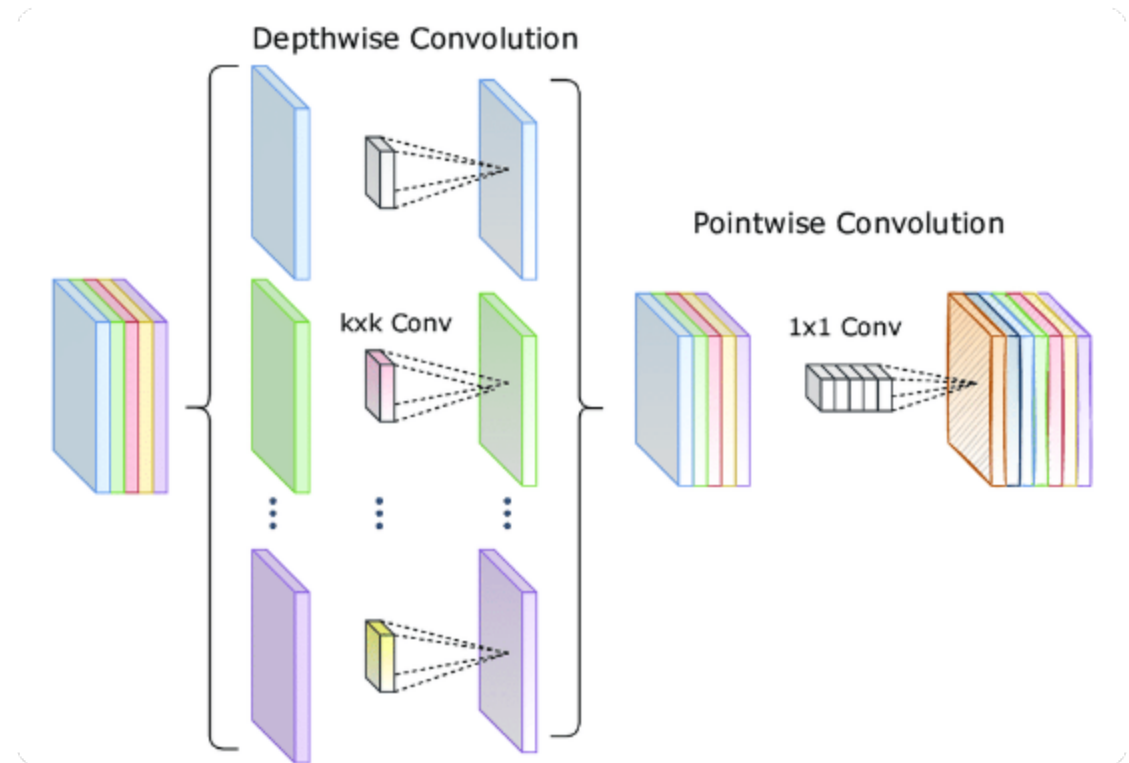
In your next practical, you will apply these concepts using a pretrained **MobileNet** model.

📁 Inspect dataset distribution.

🔗 Split into Train / Validation / Test.

🔒 **Phase 1:** Freeze MobileNet backbone. Train classifier.

🔓 **Phase 2:** Unfreeze select layers. Fine-tune gently.



Practical 6: What to Compare

Experiment	Backbone	Trainable Layers	What to Observe
A (Baseline)	MobileNet	Classifier ONLY (Dense Layer)	Initial Accuracy, Fast Training Time
B (Fine-Tuning)	MobileNet	Last few Conv layers + Classifier	Overfitting Risk, Validation Performance

Model Metrics

- Training Time
- Number of Trainable Parameters
- Signs of Overfitting

Dataset Checks

- Are the splits sensible?
- Are augmentations reasonable?
- Is there data leakage?

Recap & Next Steps

Small Data

Risks severe overfitting. Mitigate with Transfer Learning and logical Data Augmentation.

Class Imbalance

Makes standard accuracy misleading. Always inspect the Confusion Matrix and class distribution.

Dataset Splitting

Train, Validation, and Test sets must be kept strictly separate to prevent Data Leakage.

EXIT QUESTION

You have 1,000 images: 900 cats and 100 dogs. How would you prepare the dataset before training a CNN?

Next Unit: Object Detection

Image Sources



https://nsf.gov-resources.nsf.gov/styles/mobile_full_width/s3/media_hub/ImageNetCollage2-final.jpg?itok=I_LzxflW

Source: www.nsf.gov



https://media.geeksforgeeks.org/wp-content/uploads/20251213150902638109/machine_learning.webp

Source: www.geeksforgeeks.org



https://static.vecteezy.com/system/resources/thumbnails/055/522/493/small_2x/vertical-tired-frustrated-university-student-struggling-to-memorize-information-on-the-papers-trying-to-learn-class-notes-for-an-upcoming-exam-stressed-pupil-with-lack-of-motivation-camera-b-video.jpg

Source: www.vecteezy.com



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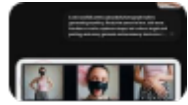
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